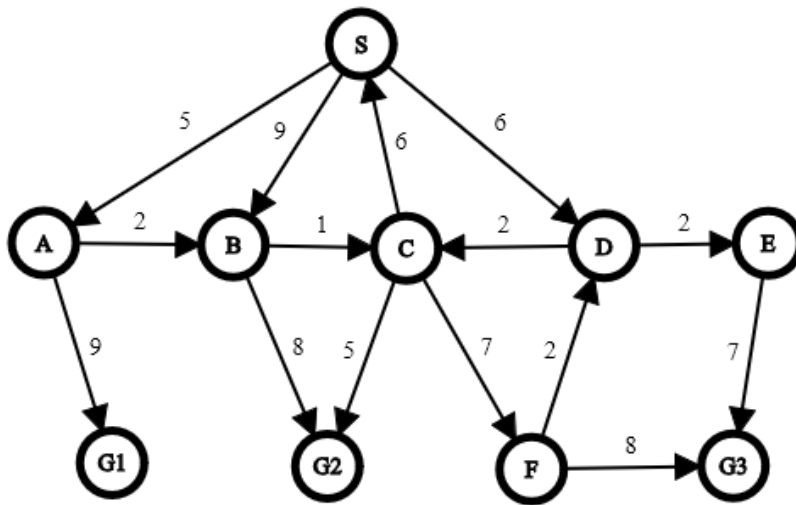


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1. Simulate A-star search on the following figure to find out **the optimal path to the goal**, as well as **the cost of the optimal path**. Heuristic values are given in the table on the right. **S** is the **starting node**, and **G1, G2, G3** are the **goal nodes**. [Hint: A-star search terminates as soon as ANY goal is found.] [8]



Node	Heuristic
S	8
A	5
B	1
C	2
D	4
E	3
F	4

Solution:

Node	g	h	f	Parent
S	0	8	8	-
A	5	5	10	S
B	9	1	10	S
D	6	4	10	S
G1	14	0	14	A
C	8	2	10	B
G2	13	0	13	B
E	8	3	11	D
F	15	4	19	C
G3	15	0	15	E

Open list: ~~S A B D~~ G1 ~~E G2~~ F G3

Closed list: S A B D C E G2

So optimal goal: G2, cost: 13

Path: S -> A -> B -> C -> G2

Note: In second iteration there was a three-way tie between A, B, and D. Here we chose A. If you choose D, you will get a different optimal path S -> D -> C -> G2, but the optimal goal (G2) and cost (13) will be the same.

2. For the figure in Q.1, is admissibility a sufficient condition to guarantee optimality? Explain in 1-2 sentences. [2]

Solution:

- For graph search of A-star, consistency is needed, admissibility is not sufficient.
- For tree search, admissibility is sufficient.

Since the figure in Q1 is a graph (it has cycles), admissibility is not sufficient to guarantee optimality, consistency is needed.

3. Imagine you have a **6×6** grid, where an agent's goal is to move from the **bottom-left corner** (0, 0) to the **top-right corner** (5, 5). The movement costs are: **horizontal or vertical = 1, diagonal = $\sqrt{2}$** . You are given TWO heuristics:

- $h_1(n)$ = **Manhattan distance** from node **n** to goal
- $h_2(n)$ = **Euclidean distance** from node **n** to goal

Using any two nodes in the grid except the source and the goal [e.g. (3,3) and (4,1)], **evaluate whether the heuristics are admissible or not.** [5]

Solution:

For (3,3): optimal path to goal is TWO diagonal moves, optimal cost $h^*(n) = 2\sqrt{2} = 2.828$

$$h_1(n) = |5 - 3| + |5 - 3| = 4 > h^*(n)$$

$$h_2(n) = \sqrt{2^2 + 2^2} = 2.828 = h^*(n)$$

For (4,1): optimal path to goal is THREE vertical moves followed by ONE diagonal move

$$\text{Optimal cost } h^*(n) = 3 + \sqrt{2} = 4.414$$

$$h_1(n) = |5 - 4| + |5 - 1| = 5 > h^*(n)$$

$$h_2(n) = \sqrt{1^2 + 4^2} = 4.17 < h^*(n)$$

Condition for admissibility: $h(n) \leq h^*(n)$

So, Manhattan distance is NOT admissible but Euclidean distance is admissible.